

### Q3 Constant acceleration equations

Name: \_\_\_\_\_

Class: \_\_\_\_\_

Date: \_\_\_\_\_

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Time: **239 minutes**

Marks: **196 marks**

Comments:

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### Q1.

A block, of mass 5 kg, is held at rest on a rough plane, which is inclined at  $30^\circ$  to the horizontal. The block is released and slides down the plane. The coefficient of friction between the block and the plane is 0.2.

- (a) Draw a diagram to show the forces acting on the block as it slides. (1)
  - (b) Show that the magnitude of the friction force acting on the block is approximately 8.5 N. (3)
  - (c) Find the acceleration of the block. (4)
  - (d) Find the speed of the block when it has travelled 1.2 metres down the slope. (3)
- (Total 11 marks)**

### Q2.

A particle is projected from a horizontal surface at a speed  $V$  and at an angle  $\alpha$  above the horizontal.

- (a) Prove that the maximum height of the particle is  $\frac{V^2 \sin^2 \alpha}{2g}$ . (6)
  - (b) A ball is hit from ground level. The ball initially moves at an angle of  $60^\circ$  above the horizontal. The maximum height of the ball is 6 metres above the ground. Modelling the ball as a particle:
    - (i) find the initial speed of the ball; (3)
    - (ii) find the range of the ball. (4)
- (Total 13 marks)**

### Q3.

A particle is initially at the origin. The particle has a constant acceleration of  $(2\mathbf{i} - 3\mathbf{j}) \text{ m s}^{-2}$ . Two seconds after it has left the origin the velocity of the particle is  $(2\mathbf{i} + 14\mathbf{j}) \text{ m s}^{-1}$ . The unit vectors  $\mathbf{i}$  and  $\mathbf{j}$  are perpendicular.

- (a) Show that the initial velocity of the particle is  $(-2\mathbf{i} + 20\mathbf{j}) \text{ m s}^{-1}$ . (3)
- (b) Find the distance of the particle from the origin after 3 seconds. (5)

- (c) The speed of the particle is  $10 \text{ m s}^{-1}$  for the first time  $T$  seconds after it has left the origin. Find  $T$ .

(6)

(Total 14 marks)

**Q4.**

A hot air balloon is released from rest at ground level and rises vertically. It accelerates at a constant rate. It reaches a height of 120 metres after 50 seconds.

- (a) Show that the acceleration of the balloon is  $0.096 \text{ ms}^{-2}$  (3)
- (b) Find the speed of the balloon when it is at a height of 200 metres. (3)
- (c) Find the time that it takes the balloon to reach a height of 200 metres. (3)
- (d) The mass of the balloon is 250 kg. Find the magnitude of the upward force acting on the balloon. (3)

(Total 12 marks)

**Q5.**

A particle moves with a constant acceleration of  $(-2\mathbf{i} - 4\mathbf{j}) \text{ m s}^{-2}$ . At time  $t = 0$  the particle is at the origin and has velocity  $(4\mathbf{i} + 7\mathbf{j}) \text{ m s}^{-1}$ . The unit vectors  $\mathbf{i}$  and  $\mathbf{j}$  are perpendicular.

- (a) Find an expression for the position vector of the particle at time  $t$  seconds. (2)
- (b) Find the time when the particle has position vector  $(3\mathbf{i} + 5\mathbf{j})$  metres. (6)

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(Total 8 marks)

**Q6.**

A van has mass 1200 kg. It travels with constant acceleration up a slope inclined at  $4^\circ$  to the horizontal. The length of the slope is 250 metres. At the bottom of the slope the van has a speed of  $20 \text{ m s}^{-1}$  and at the top its speed has dropped to  $15 \text{ m s}^{-1}$ .

- (a) Show that the acceleration of the van while it is on the slope is  $-0.35 \text{ m s}^{-1}$  (3)
- (b) A simple model assumes that no external resistance forces act on the van. A constant force of magnitude  $P$  newtons parallel to the slope acts on the van. Find  $P$ . (4)
- (c) A more realistic model assumes that a resistance force does act on the van.

This force has magnitude 300 newtons and acts parallel to the slope. Revise your answer to part (b) to take account of this extra force.

(2)

- (d) In reality the resistance force will not be constant. Explain why.

(2)

(Total 11 marks)

### Q7.

A sprinter starts from rest, and accelerates at  $2 \text{ m s}^{-2}$  for the first 4 seconds of a race. Assume that the sprinter moves along a straight line.

- (a) Find the distance travelled by the sprinter in the first 4 seconds.

(2)

- (b) Find the speed of the sprinter at the end of the first 4 seconds.

(2)

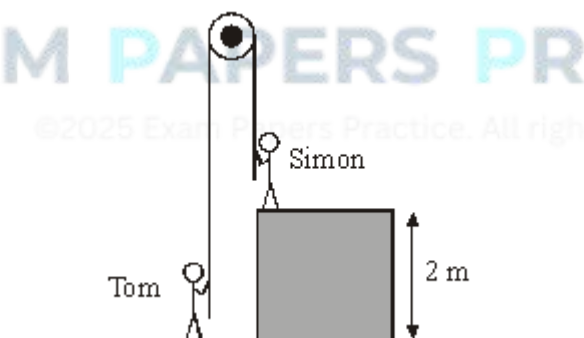
- (c) The sprinter then travels at this speed for the remainder of the race. He travels a total distance of 100 metres. Find the total time that he takes to complete the race.

(4)

(Total 8 marks)

### Q8.

Two children are holding the ends of a light, inextensible rope, which passes over a light, smooth pulley. Initially Tom, who has a mass of 40 kg, is standing at ground level and Simon, who has a mass of 60 kg, is on the edge of a fixed platform 2 metres above ground level. Model the two boys as particles, one initially at ground level, and the other initially at a height of 2 metres. The rope is taut.



Simon steps off the platform and as he falls vertically, Tom rises vertically.

- (a) Assume that the rope remains taut while the boys are moving.

- (i) Show that the acceleration of each boy is  $1.96 \text{ m s}^{-2}$ .

(5)

- (ii) Find the tension in the rope.

(2)

- (b) Find the total distance that Tom travels upwards.

(7)

(Total 14 marks)

### Q9.

A particle is projected from a horizontal surface and lands on the same horizontal surface. Its initial velocity is  $V \text{ m s}^{-1}$  at an angle  $\alpha$  above the horizontal.

- (a) Prove that the range of the particle is

$$\frac{V^2 \sin 2\alpha}{g}.$$

(6)

- (b) Hence explain why the maximum range of the particle is  $\frac{V^2}{g}$ , and state the value of  $\alpha$  for which the particle has this maximum range.

(2)

- (c) A machine for launching tennis balls fires the balls from ground level on a horizontal surface. Assume that the balls are all fired at the same speed. The maximum range of one of these balls is 35 metres.

- (i) Find the speed at which the tennis balls are fired.

(2)

- (ii) Find the range of a tennis ball fired at an angle of  $30^\circ$  above the horizontal, giving your answer to the nearest metre.

(2)

(Total 12 marks)

### Q10.

A boat moves with a constant acceleration of  $(0.2\mathbf{i} + 0.1\mathbf{j}) \text{ m s}^{-2}$ , where the unit vectors  $\mathbf{i}$  and  $\mathbf{j}$  are directed east and north respectively. Time  $t$  is measured in seconds.

At time  $t = 10$ , the boat is at the point  $A$ , which has position vector  $(30\mathbf{i} - 35\mathbf{j})$  metres.

At time  $t = 20$ , the boat is at the point  $B$ , which has position vector  $(70\mathbf{i} - 40\mathbf{j})$  metres.

- (a) Find the vector  $\overrightarrow{AB}$ .

(1)

- (b) Find the velocity of the boat when  $t = 10$ .

(3)

- (c) Using your answer to part (b), find the velocity of the boat when  $t = 0$ .

(3)

- (d) Find the position vector of the boat when  $t = 0$ .

(3)

(Total 10 marks)

### Q11.

A car accelerates uniformly along a straight horizontal road for 20 seconds. In this time its speed increases from  $8 \text{ m s}^{-1}$  to  $34 \text{ m s}^{-1}$ .

- (a) Calculate the acceleration of the car.

(3)

- (b) Find the distance that the car travels in this time.

(2)

- (c) The car has mass 1250 kg and a resistance force of magnitude 400 N acts on the car. Find the magnitude of the forward driving force that acts on the car.

(3)

(Total 8 marks)

### Q12.

A particle moves on a horizontal plane subject to the following three forces:

$$\mathbf{F}_1 = (2\mathbf{i} + 3\mathbf{j})\text{N}, \quad \mathbf{F}_2 = (6\mathbf{i} - 8\mathbf{j})\text{N} \quad \text{and} \quad \mathbf{F}_3 = (-2\mathbf{i} + 10\mathbf{j})\text{N}$$

The unit vectors  $\mathbf{i}$  and  $\mathbf{j}$  are perpendicular and lie in the horizontal plane.

Do not include any other forces in your calculations.

- (a) Find the resultant force on the particle.

(2)

- (b) The mass of the particle is 8 kg. Find the acceleration of the particle.

(1)

- (c) At time  $t = 0$ , the particle is at the origin and has velocity  $(3\mathbf{i} - 2\mathbf{j})\text{ms}^{-1}$ .

- (i) Find the velocity of the particle when  $t = 40$  seconds.

(2)

- (ii) Find the distance of the particle from the origin when  $t = 40$  seconds.

(5)

(Total 10 marks)

### Q13.

A rope is used to pull a box, of mass 20 kg, up a rough slope. Assume that the box moves with a constant acceleration.

- (a) As the box moves 4 metres up the slope its speed increases from  $1 \text{ m s}^{-1}$  to  $5 \text{ m s}^{-1}$ . Calculate the acceleration of the box.

(3)

- (b) The slope is at an angle of  $30^\circ$  to the horizontal. Assume that the rope is parallel to the slope. The coefficient of friction between the box and the slope is 0.4.

(i) Draw a diagram to show the forces on the box.

(1)

(ii) Show that the magnitude of the friction force acting on the box is approximately 68 N.

(3)

(iii) Find the tension in the rope.

(4)

(Total 11 marks)

#### Q14.

A car accelerates uniformly along a straight horizontal road. As it travels 200 metres, its speed increases from  $12 \text{ m s}^{-1}$  to  $20 \text{ m s}^{-1}$ .

(a) Show that the acceleration of the car is  $0.64 \text{ m s}^{-2}$ .

(3)

(b) Find the time that it takes the car to travel this distance.

(3)

(c) The mass of the car is 1200 kg. Using the answer from part (a), find the magnitude of the forward force that acts on the car during the above motion if:

(i) there is not resistance to its motion;

(1)

(ii) a constant resistive force of magnitude 450 newtons acts on the car.

(2)

(Total 9 marks)

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#### Q15.

A lift of mass 600 kg is suspended from a single vertical cable. The lift is initially at rest and then starts to accelerate upwards. The tension in the cable remains constant at 6060 newtons while the lift is accelerating. Assume that there is no resistance to the motion of the lift.

(a) Show that the acceleration of the lift is  $0.3 \text{ m s}^{-2}$ .

(2)

(b) Find the time that it would take the lift to travel 8 metres.

(3)

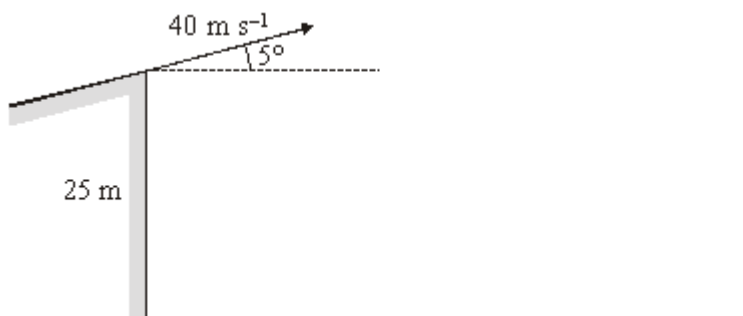
(c) Find the speed of the lift when it has travelled 8 metres.

(2)

(Total 7 marks)

**Q16.**

In a film a stunt-man drives a car off the top of a vertical cliff. The top of the cliff is 25 metres above the level of the sea. When it leaves the cliff the car is travelling at  $40 \text{ m s}^{-1}$  at an angle of  $5^\circ$  above the horizontal. The diagram shows the cliff and the initial velocity of the car.



Model the car as a particle and assume that, while it is in the air, it moves under the influence of gravity alone.

- (a) Show that the car hits the sea approximately 2.64 seconds after it leaves the top of the cliff.

(6)

- (b) Find the horizontal distance of the car from the cliff when the car hits the sea.

(2)

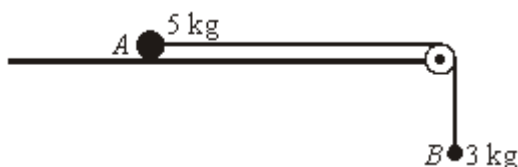
- (c) Find the speed of the car when it hits the sea.

(5)

(Total 13 marks)

**Q17.**

The diagram shows two particles that are connected by a light inextensible string. The 5 kg particle,  $A$ , is on a rough horizontal surface and the 3 kg particle,  $B$ , hangs on the other end of the string as shown in the diagram. The string passes over a smooth light pulley.



The coefficient of friction between  $A$  and the horizontal surface is 0.8. The particles are set in motion with the string taut and  $A$  moving towards the pulley. The initial speed of both particles is  $2 \text{ m s}^{-1}$ . The particles slow down and stop before  $A$  reaches the pulley or  $B$  hits the floor.

- (a) Show that, while both particles are moving, the magnitude of their acceleration is  $1.225 \text{ m s}^{-2}$ .

(6)



- (b) Find the tension in the string while the particles are moving.

(2)

- (c) Find the distance that the particles move before they come to rest.

(3)

(Total 11 marks)

**Q18.**

At time  $t = 0$ , a particle is at the origin and moving with velocity  $(4\mathbf{i} + 2\mathbf{j}) \text{ m s}^{-1}$ .  
When  $t = 10$  seconds the position vector of the particle is  $(44\mathbf{i} + 28\mathbf{j})$  metres.  
The particle is moving with constant acceleration.

- (a) Find the acceleration of the particle.

(4)

- (b) Find the position vector of the particle at time  $t$ .

(3)

- (c) At time  $t = T$ , the position vector of the particle is  $(96\mathbf{i} + 72\mathbf{j})$  metres.

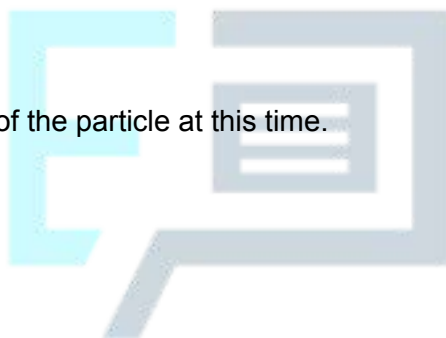
- (i) Find  $T$ .

(4)

- (ii) Find the speed of the particle at this time.

(3)

(Total 14 marks)



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